

Nikita Kalinin

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Interests: analytic/elementary number theory around $SL(2, \mathbb{Z})$, Farey fractions, topographs, and lattice sums; continued fractions, affine geometry.

Positions

Associate professor (tenure track), *Guangdong Technion-Israel Institute of Technology*, Shantou, China (01.2024 – Present).

Visiting Associate professor, *Guangdong Technion-Israel Institute of Technology*, Shantou, China (10.2022 – 11.2023).

Associate professor (доцент), *Department of Mathematics and Computer Science*, Saint Petersburg University, Russia (09.2019 – 11.2022).

Senior Researcher (старший научный сотрудник), *International Laboratory of Game Theory and Decision Making, Saint Petersburg branch of HSE* (National Research University Higher School of Economics), Russia (09.2017 – 10.2022).

A holder of a Swiss grant PostDoc.Mobility with Ernesto Lupercio as a hosting professor, *CINVESTAV* (*Centro de Investigación y de Estudios Avanzados*), Mexico City, Mexico (07.2016 – 08.2017).

Education

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| 2015 | PhD in Mathematics, <i>University of Geneva</i> , Switzerland
Thesis topic: “Tropical geometry for Nagata’s conjecture and Legendrian curves”
Advisor: Grigory Mikhalkin |
| 2010 | Specialist (=master) in Mathematics, <i>Saint Petersburg University</i> , Russia
Thesis topic: “The Alexander polynomial as an intersection of two cycles in a symmetric power”
Advisor: Oleg Ya. Viro |

Honors, Awards, and Grants

2026	Technion-GTIIT seed grant, Probabilistic methods in geometry (joint with Ilya Gekhtman)
2020	Russian Science Foundation grant №20-71-00007, Scaling in sandpile models, 07.2020–06.2022
2018	Young Russian Mathematics grant holder, A summation over subsets of $SL(2, \mathbb{Z})$, 01.2018–12.2020
2016	PostDoc.Mobility SNSF (Switzerland) grant, P2GEP2_168647, Sandpiles in physics, combinatorics, tropical and algebraic geometry 1.07.2016 – 31.08.2017
2005	Gold medal at the International Mathematical Olympiad

Service

2020-2024	Editor-in-chief of a book about St. Petersburg mathematicians, <i>Saint Petersburg mathematicians and their discoveries</i> , 496p., (2024), MCCME. Freely available on https://sites.google.com/view/spbmth , Russian version is available on https://biblio.mccme.ru/node/291128
2020	Senior Coordinator on International Mathematical Olympiad 2020

Postdocs

2025–...	Faith Shadow Zottor
2026–...	Abdul Quadir

Teaching Experience

GTIIT:	<i>Numerical analysis,</i> <i>Set theory,</i> <i>Algebra B,</i> <i>Basic notions of mathematics,</i> <i>Infinitesimal analysis 3</i> <i>Combinatorial geometry</i> <i>Seminar in Geometry</i> <i>Probability theory</i> <i>Number theory</i>
SPbU:	<i>Symplectic topology,</i> <i>Introduction to homotopy theory,</i> <i>Game theory,</i> <i>Introduction to homology theory,</i> <i>General topology,</i> <i>Hyperbolic geometry,</i> <i>Topological data analysis</i> <i>Differential forms in algebraic topology</i>
HSE:	<i>Algebra and analysis</i> (for sociologists), <i>Game theory</i> (for economists), <i>Topological data analysis</i> (for programmers)
UniGe:	<i>Complex Analysis</i> (assistant; in French), <i>Characteristic classes</i> (assistant; in English), <i>Méthodes élémentaires</i> (in French)

Master theses under my supervision

2024	Triangulations of flat $\mathbb{R}P^2$ with three different conical points
2024	Sandpile models on oriented graphs
2017	Sandpile phenomena in tropical systems
2014	Un polynôme pour les matroïdes signés: généralisations et applications

Bachelor theses under my supervision

2024	Sandpile on infinite graphs
2023	Newton polytopes of singular surfaces
2023	Triangulations of flat $\mathbb{R}P^2$ with four conical points
2023	Densities of lattices of translates
2023	Patterns in sandpiles
2023	Growth phenomena in sandpiles
2022	Analysis of the mechanism of the college admission in Russia in 2021
2021	Equilibrium existence in the case of non-divisible objects
2020	Modeling and analysis of crab auction in Russia
2020	Repeated auctions with a reserve price
2019	Isoperimetric problem for a family of closed curves on a sphere

Preprints

Many-point tropical relaxation and the Monge–Ampère equation (joint with Ernesto Lupercio, Higinio Serrano, Mikhail Shkolnikov), <https://arxiv.org/abs/2607.25878>

The directed Abelian sandpile model on cylinders (joint with Abdul Quadir, Ram Ramaswamy), <http://arxiv.org/abs/2605.15914>

Residues of a tropical zeta function for convex domains (joint with Ernesto Lupercio, Mikhail Shkolnikov), <https://arxiv.org/abs/2604.21709>

Error terms for continued fractions of $e^{1/s}$ and $\sqrt{\frac{v}{u}} \tanh\left(\frac{1}{\sqrt{uv}}\right)$ (joint with Takao Komatsu), <https://arxiv.org/abs/2604.09726>

CayleyPy-4: AI-Holography. Towards analogs of holographic string dualities for AI tasks (joint with other 43 authors), <https://arxiv.org/abs/2603.22195>

A p -adic ($p \equiv 3 \pmod{4}$) depth-5 supercongruence for Gaussian p -th power sums over a square, (joint with F.S. Zottor) <http://arxiv.org/abs/2602.00206>

Evaluation of lattice sums via telescoping over topographs, <https://arxiv.org/abs/2510.02082>

Legendre duality for certain summations over the Farey pairs, <https://arxiv.org/abs/2409.10592>

Sandpile group of infinite graphs (joint with V. Khramov), <https://arxiv.org/abs/2305.05346>

Shrinking dynamic on multidimensional tropical series, <https://arxiv.org/abs/2201.07982>

Published papers

Tropical curves in sandpile models, **Advances in Mathematics** 502 (2026) 111127, (joint with M. Shkolnikov)

Topology of a uniform spanning tree on a cylinder, **Physica Scripta**, Volume 101, 325204 (2026), (joint with D. Rakhmankin)

Limits of equi-affine equi-distant loci of planar convex domains with two non-parallel asymptotes, **SIGMA** 22 (2026), 071, 10 pages, (joint with M. Shkolnikov)

Weighted error-sum identities for periodic continued fractions and their generalizations, (joint with K. Calderon) <https://arxiv.org/abs/2601.07862>

A half-shift reflection identity for the digamma function, **Research in Number Theory**, 12, 75, (2026)

A guide to tropical modifications, **Serdica Mathematical Journal** 52 (2), 109-156 (2026)

Proof of Sun's conjectures on hyperbolic cosine series via the Eisenstein–Lambert method, **Frontiers in Combinatorics and Number Theory**, Volume 3, pp.21–36 (2026)

Evaluating lattice sums via telescoping on $SL_+(2, \mathbb{Z})$: a short proof of $\sum \frac{1}{\|x\|^2 \|y\|^2 \|x+y\|^2} = \frac{\pi}{4}$ and of Zagier's identity, **International Journal of Number Theory** (2026)

Continued fractions, determinant expressions, and identities, **AIMS Mathematics** Volume 11, Issue 4: 9712-9735 (2026), (joint with Takao Komatsu)

A short telescoping proof of Hata's formula for the Euler–Mascheroni constant, **Integers**, 25, A90 (2025)

Wolstenholme's theorem over Gaussian integers, **Funct. Approx. Comment. Math.** 74.2 (2026), pp.213-223

Tropical Weil's reciprocity law and Weil's pairing, **Journal of Mathematical Sciences**, (joint with M. Magin)

(Sandpile patterns on a regular graph of degree eight, in Russian, joint with P. Granin, A. Saakyan)
Песочные паттерны на регулярном графе с вершинами степени восемь, **Chebyshevskii Sbornik**, vol. 25 (2024), no. 3, pp.47–69

(Convergence of a sandpile on a triangular lattice under rescaling, in Russian, joint with A. Aliev)
Сходимость песочной кучи на треугольной решётке при ремасштабировании, **Matematicheskii Sbornik**, vol. 214 (2023), issue 12, pp.3–25

Some statistics about Tropical Sandpile Model, **Communications in Mathematics**, Volume 31 (2023), Issue 3, 9–19 (joint with Y. Prieto)

Sandpile solitons in higher dimensions, **Arnold Mathematical Journal**, 9 (2023), no. 3, 435–454

Equilateral convex triangulations of $\mathbb{R}P^2$ with three conical points of equal defect, **In the Tradition of Thurston II: Geometry and Groups** (2022, p. 315–329), editors: Ken'ichi Ohshika, Athanase

Papadopoulos, Springer (joint with M. Chernavskikh, A. Erdnigor, and A. Zakharov)

Sandpile Solitons via Smoothing of Superharmonic Functions, **Communications in Mathematical Physics**, 378(3) (2020), 1649–1675 (joint with M. Shkolnikov)

Pattern Formation and Tropical Geometry, **Frontiers in Physics**, 2020 (8), 423

Strategic analysis of the Russian crab quota auction in 2019, **Marine Policy**, Volume 122, December 2020, 104266, (joint with Mark Vershinin)

Tropical formulae for summation over a part of $SL(2, \mathbb{Z})$, **European Journal of Mathematics**, Volume 5 (2019), Issue 3, 909–928 (joint with M. Shkolnikov)

Self-Organized Criticality and Pattern Emergence through the Lens of Tropical Geometry, **PNAS** August 28, 2018. 115 (35) E8135–E8142 (joint with A. Guzmán-Sáenz, Y. Prieto, M. Shkolnikov, V. Kalinina, E. Lupercio)

Legendrian curves in $\mathbb{C}P^3$: cubics and curves on a quadratic surface, **Zapiski Nauchnyh Seminarov PDMI**, vol. 476 (2018), Geometry and Topology Series 13, 92–110 (Journal of Mathematical Sciences, 251(4), 489–502)

Introduction to tropical series and wave dynamic on them, **Discrete and continuous dynamical systems**, **A**, 38(6) (2018): 2843–2865 (joint with M. Shkolnikov)

The number π and summation by $SL(2, \mathbb{Z})$, **Arnold Mathematical Journal**, Volume 3 (2018), Issue 4, 511–517 (joint with M. Shkolnikov)

Tropical approach to Nagata’s conjecture in positive characteristic, **Discrete and Computational Geometry**, 58(1) (2017), 158–179

Tropical curves in sandpiles, **Comptes Rendus Mathématique**, Volume 354, Issue 2, 1 February 2016, 125–130 (joint with M. Shkolnikov)

Sandpiles on the heptagonal tiling, **Knot Theory and Its Ramifications**, Vol. 25 (2016), Issue 12, 1642005 (joint with M. Shkolnikov)

The Newton polygon of a planar singular curve and its subdivision, **Journal of Combinatorial Theory, Series A**, 137 (2016), 226–256

The Alexander polynomial as an intersection of two cycles in a symmetric power, **Journal of Knot Theory and Its Ramifications**, Vol. 24 (2015), No. 12, 1550061

Articles in Russian

(What could be a dynamical centralized admission system in Russia, in Russian, **Journal of the New Economic Association**) Каким могло бы быть централизованное распределение абитуриентов по образовательным программам, **Журнал Новой Экономической Ассоциации**, № 1 (62), 2024, 101–115 (joint with A. Kuzmina)

(A cipher broken by Ch. Goldbach, in Russian) Шифр, который разгадал Христиан Гольдбах, **Математика в высшем образовании**, 19, с. 85–94, 2021

(Crab auction in 2019: history, estimates and alternative scenarios, in Russian) Крабовый аукцион 2019: история, оценка и альтернативные сценарии, **Экономический журнал ВШЭ**, 2021. Т. 25. № 4. С. 574–594 (joint with D. Ivanov, A. Nesterov, I. Susin)

(Various equilibria in a game-theoretical model of the behavior of a teacher and a student (possibly) cheating on a written exam, in Russian) Различные равновесия в теоретико-игровой модели поведения преподавателя и (возможно) списывающего студента на письменном экзамене, **Социология: методология, методы, математическое моделирование (Социология: 4М)**, вып. 50-51, стр. 97–140

Recent conference talks

28.03.2026	“Zeta-series for convex domains”, COmbinatorial Number Theory And Connected Topics – V, Ahmedabad University, India
25.09.2025	“Statistics of Tropical Sandpiles: Degree Growth, Coefficient Laws, and Tree Structure of Curves”, Recent Developments on Tropical Sandpiles and Related Subjects, Sofia (ICMS-Sofia), Bulgaria
7.07.2025	“The residue of a zeta-function for convex domains”, The 9th National Conference on Theory of Spaces of Holomorphic Functions, Hulunbuir, China
21.01.2025	“Zeta-functions for convex domains”, MIST: Mathematics Inspired by String Theory 2025, Hong Kong, 21-24 January 2025
06.01.2025	“Tropical differential forms on tropical curves”, Chinese conference on complex analysis, Shenzhen

Recent talks on seminars

6.08.2026	“A strange divisibility by two in the sandpile group”, MIPT, Moscow
8.06.2026	“Chebyshev’s 1857 paper: When an Elliptic-Looking Integral Becomes Elementary”, Shenzhen MSU-BIT University
24.04.2026	“Evaluation of lattice sums via telescoping over topographs”, Algebra and Combinatorics Seminar, The University of Texas at Dallas
19.02.2026	“Evaluation of lattice sums via telescoping over topographs”, Special Functions and Number Theory seminar, India
15.01.2026	“Sandpile and tropical sandpile models”, YMSC Probability seminar, Tsinghua University, Beijing, China
28.11.2025	“Telescoping over topograph”, Colloquium Saint-Peterburg University, Russia
5.11.2025	“Tropical Weil Reciprocity”, Moscow-Beijing Topology Seminar (online)
3.07.2025	“Zeta function for convex domains”, Harbin Polytechnical University, Harbin, China
16.06.2025	“Sandpile States on a Cylinder and Random Spanning Trees”, Jimei University, Xiamen, China
30.04.2025	“Three Lectures on Binary Quadratic Forms and Conway’s Topographs”, Sofia, Bulgaria
28.01.2025	“Elementary $SL(2, \mathbb{Z})$ number theory formulas which stem from the sandpile model”, Stavanger, Norway
17.11.2024	“Number theory in olympiads and scientific life”, Harbin Polytechnical University, Harbin, China
20.09.2024	“Tropical formulae for the number π ”, University of Geneva
6.08.2024	“Several formulae for summation over $SL(2, \mathbb{Z})$ ”, SYSU, Sun Yat-sen University, Guangzhou
21.03.2024	“Sandpiles on infinite domains”, Westlake Math Colloquium, Westlake University, Hangzhou